Jaechul Roh

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EDUCATION

University of Massachusetts Amherst — **September 2023 – May 2028 (expected)**
Ph.D. Candidate in Computer Science — Amherst, Massachusetts, USA
Advisor: Prof. Amir Houmansadr
GPA: 4.0/4.0

University of Massachusetts Amherst — **September 2023 – May 2026**
M.S. in Computer Science — Amherst, Massachusetts, USA
Advisor: Prof. Amir Houmansadr
GPA: 4.0/4.0

Hong Kong University of Science and Technology — **September 2017 – May 2023**
B.Eng. in Computer Engineering, School of Engineering — Clear Water Bay, Hong Kong
Final Year Thesis Advisor: Prof. Jun Zhang
**2 years of Compulsory Korean Military Duty*

RESEARCH INTERESTS

My research focuses on the **Safety & Security of AI models and agents**. Under the supervision of Prof. Amir Houmansadr, I investigate modality-specific threats and vulnerabilities of multimodal generative and agentic systems across audio, text, and vision, with work published at NeurIPS, USENIX Security, and COLM. I have also collaborated with researchers at Google DeepMind and Qualcomm. Most recently, I completed a research internship at Dolby Laboratories, where I worked on the safety and security of real-time full-duplex speech models.

WORK EXPERIENCE

Dolby Laboratories — **June 2026 – September 2026**
Ph.D. Research Intern, ***Supervisor****: Deepak Chandran, Andrea Fanelli* — San Francisco, California

- Investigated interruption-induced safety failures in real-time full-duplex speech models.

Brave Software — **June 2025 – September 2025**
Ph.D. Research Intern, ***Supervisor****: Ali Shahin Shamsabadi* — London, United Kingdom (Remote)

- Investigated privacy leakage and sensitive information oversharing in autonomous web agents.

ON-GOING RESEARCH

Multi-Agent Persuasion Through Speech

- Investigating speech-based persuasion attacks on AI agents, including model extraction, policy enforcement, and cybersecurity gatekeeper scenarios.
- Evaluating how persuasive arguments and vocal cues affect agent decisions and adherence to security policies.

Speaker Authorization in Voice-Driven AI Agents

- Investigating how unauthorized speakers redirect AI agents' tool calls in multi-speaker interactions.
- Evaluating speaker verification and turn-level attribution for access control in voice-driven agents.

SELECTED PUBLICATIONS

1. **DuplexJail: Spoken Interruption Attacks on Full-Duplex Speech Models**
   Jaechul Roh, Deepak Chandran, Amir Houmansadr, Andrea Fanelli
   Preprint at arXiv (Dolby Laboratories · Internship)
   [paper] [demo]

2. `CodecAttack`**: Codec-Robust Latent-Space Prompt-Injection Attacks on Audio LLMs**
   Jaechul Roh, Jean-Philippe Monteuuis, Jonathan Petit, Amir Houmansadr
   Preprint at arXiv (Qualcomm · Research Collaboration)
   [paper] [code]

3. **SPILLage: Agentic Oversharing on the Web**
   Jaechul Roh, Eugene Bagdasarian, Hamed Haddadi, Ali Shain Shamsabadi
   Preprint at arXiv (Brave Software · Internship)
   [paper] [code] [article]

4. **Benign Fine-Tuning Breaks Safety Alignment in Audio LLMs**
Jaechul Roh, Virat Shejwalkar, Amir Houmansadr
Preprint at arXiv (Google DeepMind · Research Collaboration)
[paper] [code]

5. **Multilingual and Multi-Accent Jailbreaking of Audio LLMs**
Jaechul Roh, Virat Shejwalkar, Amir Houmansadr
COLM 2025 (Google DeepMind · Research Collaboration)
[paper] [code]

6. **Backdooring Bias (B^2) into Stable Diffusion Models**
Ali Naseh, **Jaechul Roh**, Eugene Bagdasarian, Amir Houmansadr
USENIX Security '25
[paper] [code]

7. **OSLO: One-Shot Label-Only Membership Inference Attacks**
Yuefeng Peng, **Jaechul Roh**, Subhransu Maji, Amir Houmansadr
NeurIPS 2024
[paper]

8. ***Bob's Confetti*: Phonetic Memorization Attacks in Music and Video Generation**
Jaechul Roh, Zachary Novack, Yuefeng Peng, Niloofar Mireshghallah, Taylor Berg-Kirkpatrick, Amir Houmansadr
Preprint at arXiv
[paper] [demo page] [video]

9. **OverThink: Slowdown Attacks on Reasoning LLMs**
Abhinav Kumar, **Jaechul Roh**, Ali Naseh, Marzna Karpinska, Mohit Iyyer, Amir Houmansadr, and Eugene Bagdasarian
Preprint at arXiv
[paper] [code]

INVITED TALKS

1. **AI Safety Night SF** **August 2026**
Invited Talk San Francisco
   - Provided a talk on modality-unique safety and security.
2. **Google Speech Technologies Group** **July 2025**
Invited Talk Google DeepMind
   - Presented our COLM paper *"Multilingual and Multi-Accent Jailbreaking of Audio LLMs"* to the NFM Reading Group led by the Speech Technologies Group at Google DeepMind.

HONORS & AWARDS

2026 Google PhD Fellowship **April 2026**
UMass Amherst Nominee
- Selected as departmental representative candidate for the fellowship

2026 Apple Scholars in AI/ML PhD Fellowship **September 2025**
UMass Amherst Nominee
- Selected as departmental representative candidate for the fellowship

ACADEMIC SERVICES

Program Committee Member (Reviewer)

- **2027**: *AAAI, ICLR, ICASSP*
- **2026**: *ICLR, ICASSP, ICML (**Best Reviewer**), COLM, NeurIPS*
- **2025**: ICLR, COLM XLLM-Workshop

SKILLS / LANGUAGES

Programming Language: Python, C++
Languages: Korean (Native), English (Native), Chinese (Fluent)